European Commission

Operational Forest Degradation Monitoring in the Tropics

Introducing the Forest Canopy Disturbance Monitoring (FCDM) Tool

2nd EEBIOMASS WORKSHOP

15.06.2021

Andreas Langner

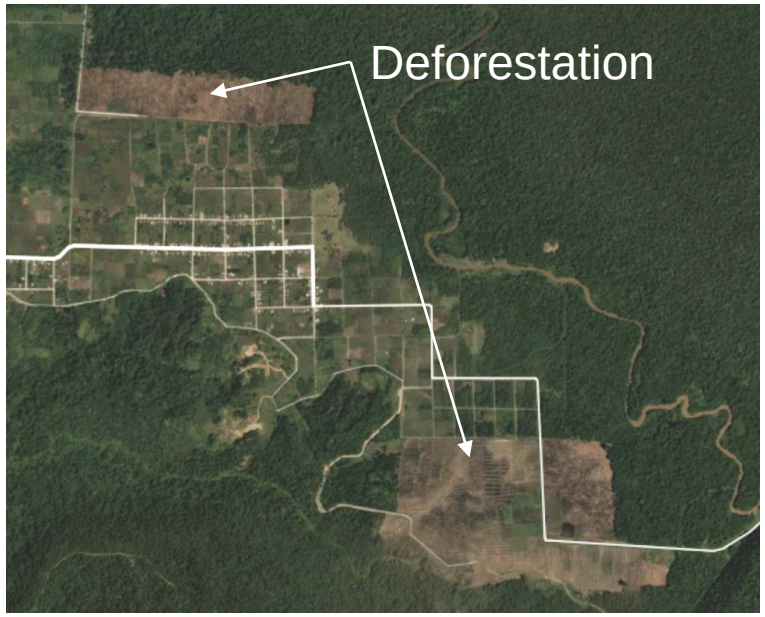
JRC.D1

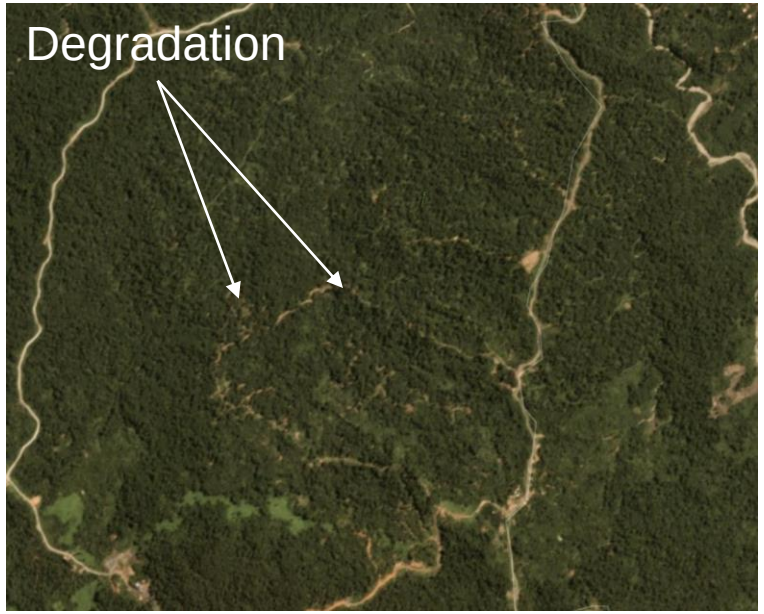
Joint Research Centre

Challenges

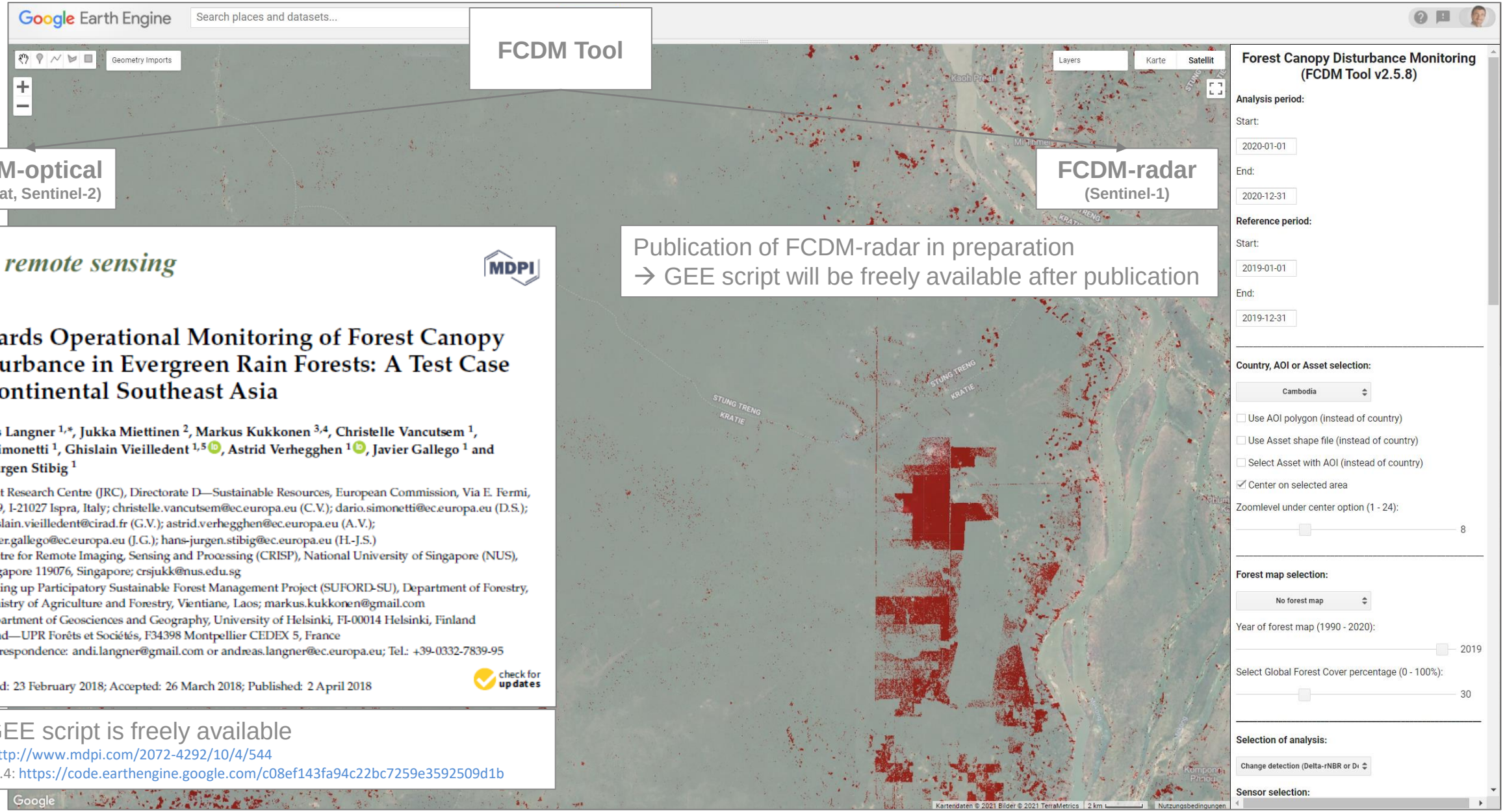
Problems to detect forest disturbances/ degradation

- Small-scale signal (e.g. removal of single trees)
- Weak signal (low signal-to-noise ratio)
- Signal only detectable over a short period due to
  - fast vegetation regrowth in tropics
  - frequent cloud coverage in tropics
- Differentiation between natural phenological changes (e.g. seasonality) and actual disturbance events





GUI of FCDM Tool in Google Earth Engine



FCDM Tool

FCDM-optical
(Landsat, Sentinel-2)

FCDM-radar
(Sentinel-1)

Publication of FCDM-radar in preparation
→ GEE script will be freely available after publication

remote sensing

MDPI

Article

Towards Operational Monitoring of Forest Canopy Disturbance in Evergreen Rain Forests: A Test Case in Continental Southeast Asia

Andreas Langner [1,*], Jukka Miettinen [2], Markus Kukkonen [3,4], Christelle Vancutsem [1], Dario Simonetti [1], Ghislain Vieilledent [1,5], Astrid Verhegghen [1], Javier Gallego [1] and Hans-Jürgen Stibig [1]

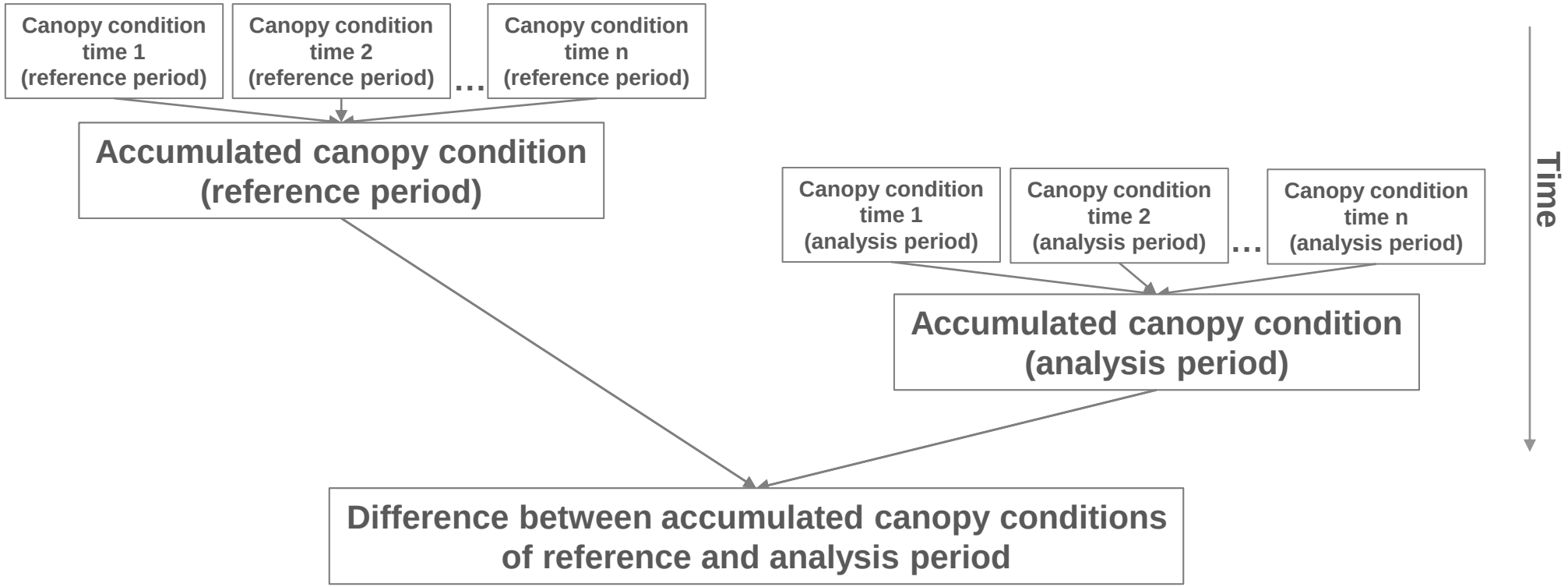
1 Joint Research Centre (JRC), Directorate D—Sustainable Resources, European Commission, Via E. Fermi, 2749, I-21027 Ispra, Italy; christelle.vancutsem@ec.europa.eu (C.V.); dario.simonetti@ec.europa.eu (D.S.); ghislain.vieilledent@cirad.fr (G.V.); astrid.verhegghen@ec.europa.eu (A.V.); javier.gallego@ec.europa.eu (J.G.); hans-jurgen.stibig@ec.europa.eu (H.-J.S.)
2 Centre for Remote Imaging, Sensing and Processing (CRISP), National University of Singapore (NUS), Singapore 119076, Singapore; crsjukk@nus.edu.sg
3 Scaling up Participatory Sustainable Forest Management Project (SUFORD-SU), Department of Forestry, Ministry of Agriculture and Forestry, Vientiane, Laos; markus.kukkonen@gmail.com
4 Department of Geosciences and Geography, University of Helsinki, FI-00014 Helsinki, Finland
5 Cirad—UPR Forêts et Sociétés, F34398 Montpellier CEDEX 5, France
* Correspondence: andi.langner@gmail.com or andreas.langner@ec.europa.eu; Tel.: +39-0332-7839-95

Received: 23 February 2018; Accepted: 26 March 2018; Published: 2 April 2018

check for updates

→ GEE script is freely available
Paper: http://www.mdpi.com/2072-4292/10/4/544
Script v2.4: https://code.earthengine.google.com/c08ef143fa94c22bc7259e3592509d1b

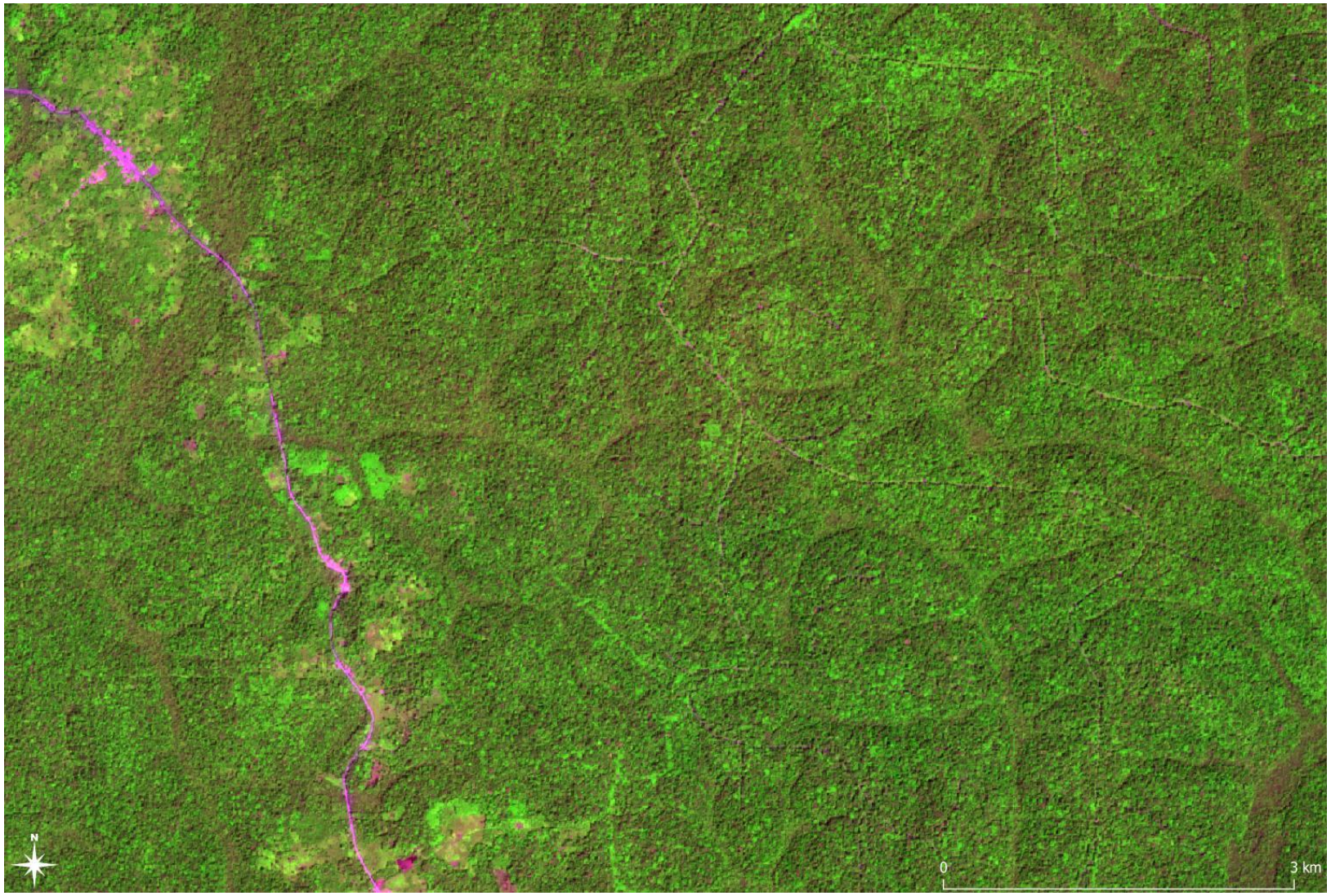
Basic Concept of the two FCDM Methodologies



	Reference period	Analysis period	FCDM result
Canopy Opening	No	Yes	Disturbance
	Yes	No	Regrowth*
	Yes	Yes	No change
	No	No	No change

European Commission

Comparison of Different Monitoring Approaches

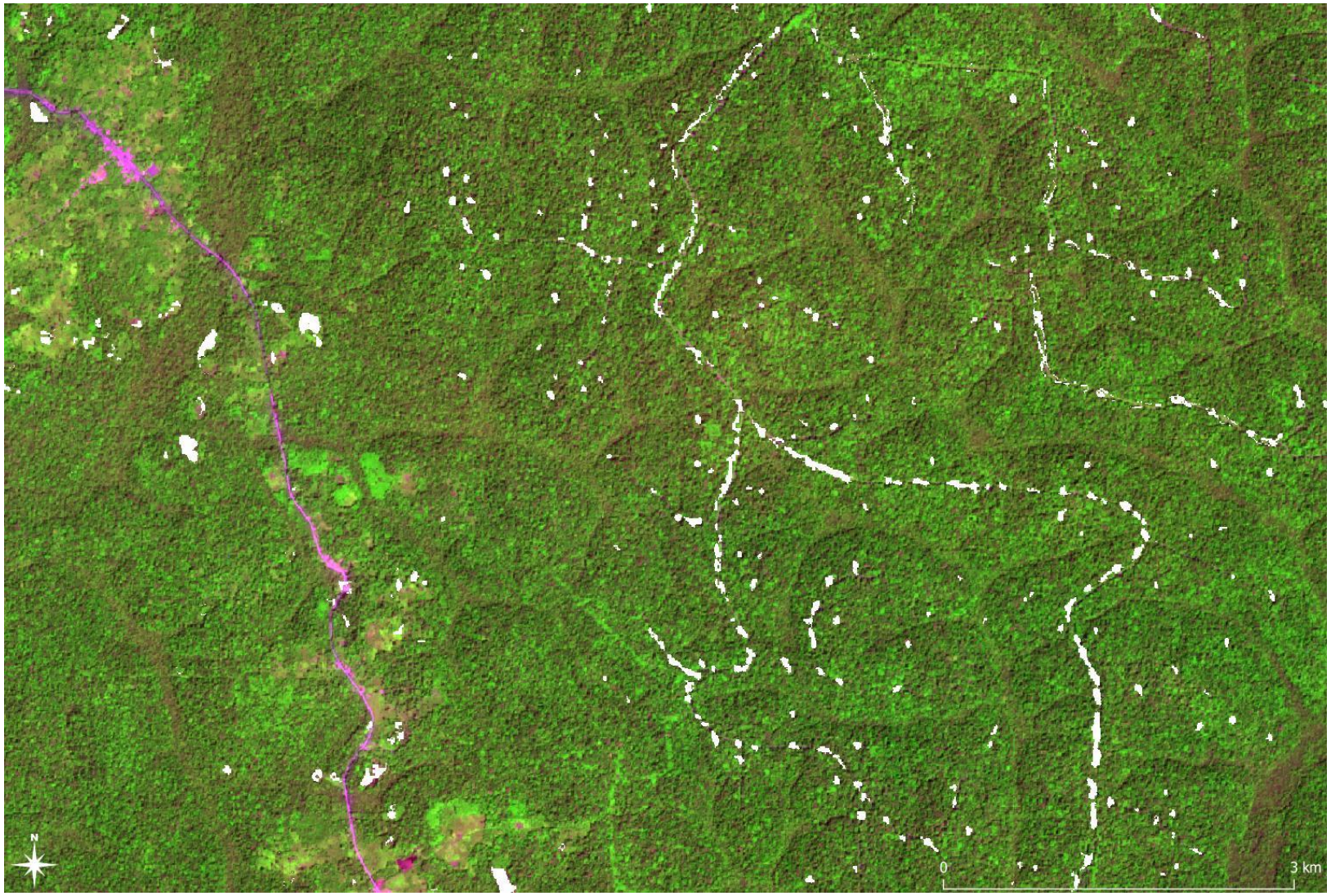


Cameroon:
Analysis period:
01.01.2020 – 31.12.2020

- Planet data (01.12.2020)

European Commission

Comparison of Different Monitoring Approaches

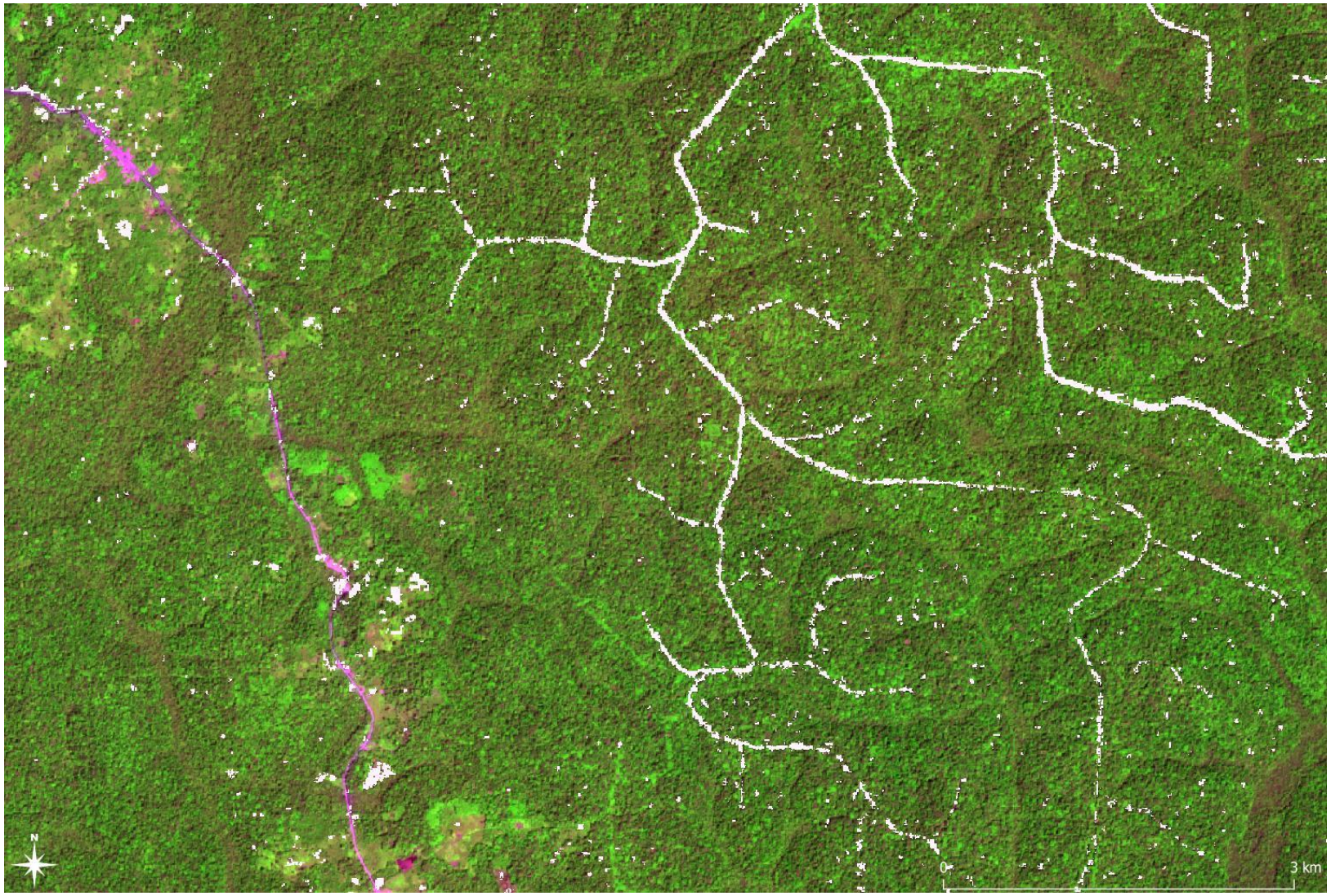


Cameroon:
Analysis period:
01.01.2020 – 31.12.2020

- Planet data (01.12.2020)
- RADD alerts 2020

European Commission

Comparison of Different Monitoring Approaches



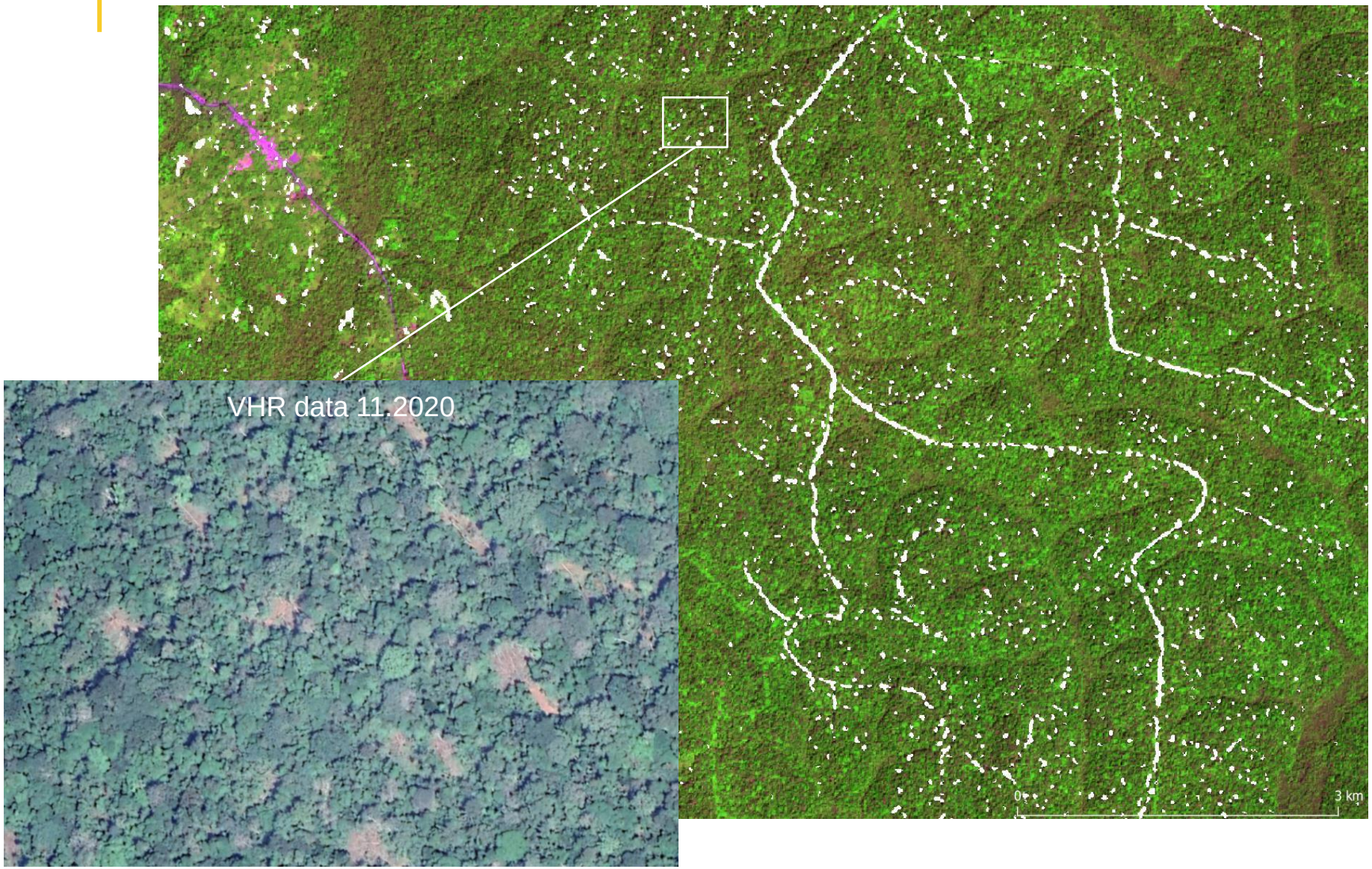
Cameroon:
Analysis period:
01.01.2020 – 31.12.2020

- Planet data (01.12.2020)
- RADD alerts 2020

- FCDM-optical (S2) 2020

European Commission

Comparison of Different Monitoring Approaches



Cameroon:
Analysis period:
01.01.2020 – 31.12.2020

- Planet data (01.12.2020)
- RADD alerts 2020

- FCDM-optical (S2) 2020
- FCDM-radar 2020

European Commission

Conclusion

FCDM approaches have capacity to detect forest disturbances/ degradation

- Change detection approaches are sensitive to small-scale and/ or weak signal (e.g. removal of single trees)
- Change detection approaches analyse every available image over analysis period → detecting even short-term disturbances
- Radar-based approach allows monitoring independent of cloud coverage
- Radar-based approach allows differentiation between seasonal changes and actual disturbance events